Final Project : Financial News Reporting System

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Instructions

1. All screenshots should be clear, high-quality, and legible, ensuring that details are easily visible. Wherever applicable, the learner's account details should be visible to verify ownership and authenticity.

2. Red boxes should be correctly placed to highlight the key steps in each screenshot, ensuring clarity and guiding the viewer through the process.

3. The overall structure and logical flow of the process must be maintained by placing each screenshot under the appropriate slide headers. Learners should avoid skipping steps or presenting them in an incorrect order, as this can disrupt the coherence of the submission.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Step 1 : Define Business context and objective

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

1.1 Business Context

FinArk operates in a high-stakes, data-driven financial services environment where timely access to accurate market intelligence is critical for investment decisions, risk management, and competitive positioning.

Today's financial news ecosystem spans written articles, earnings call recordings, analyst briefings, and visual data such as technical charts. This creates a constant stream of heterogeneous, unstructured information that is difficult to monitor and interpret manually.

To remain competitive, FinArk needs an intelligent system that can continuously ingest, interpret, and structure this multimodal financial news into actionable insights.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

1.2 Problem Statement

FinArk currently struggles to efficiently process large volumes of heterogeneous financial news data—including text articles, audio recordings, and visual information—in real time.

Manual and semi-automated workflows cannot keep up with the volume, variety, and speed of incoming information, leading to:

- Information overload for analysts
- Fragmented insights across different data modalities
- Delays in identifying market-moving events
- Limited transparency and standardization for compliance and risk teams

As a result, critical opportunities may be missed and risk exposure may increase.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

1.3 Objective

The objective is to design a Financial News Intelligence System that leverages Large Language Models (LLMs) and multimodal AI on Microsoft Azure to automate the end-to-end processing of financial news.

The system will:
- Aggregate and process text articles and audio recordings (and conceptually extend to visual data).
- Convert audio into text using speech-to-text capabilities.
- Classify each item by Sentiment (Positive/Negative/Neutral), Sector (e.g., Technology, Healthcare, Energy), and Impact Level (High/Medium/Low).
- Generate concise, structured financial reports for each article and audio file.

This enables faster, more consistent, and auditable decision-making.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

1.4 Data Description

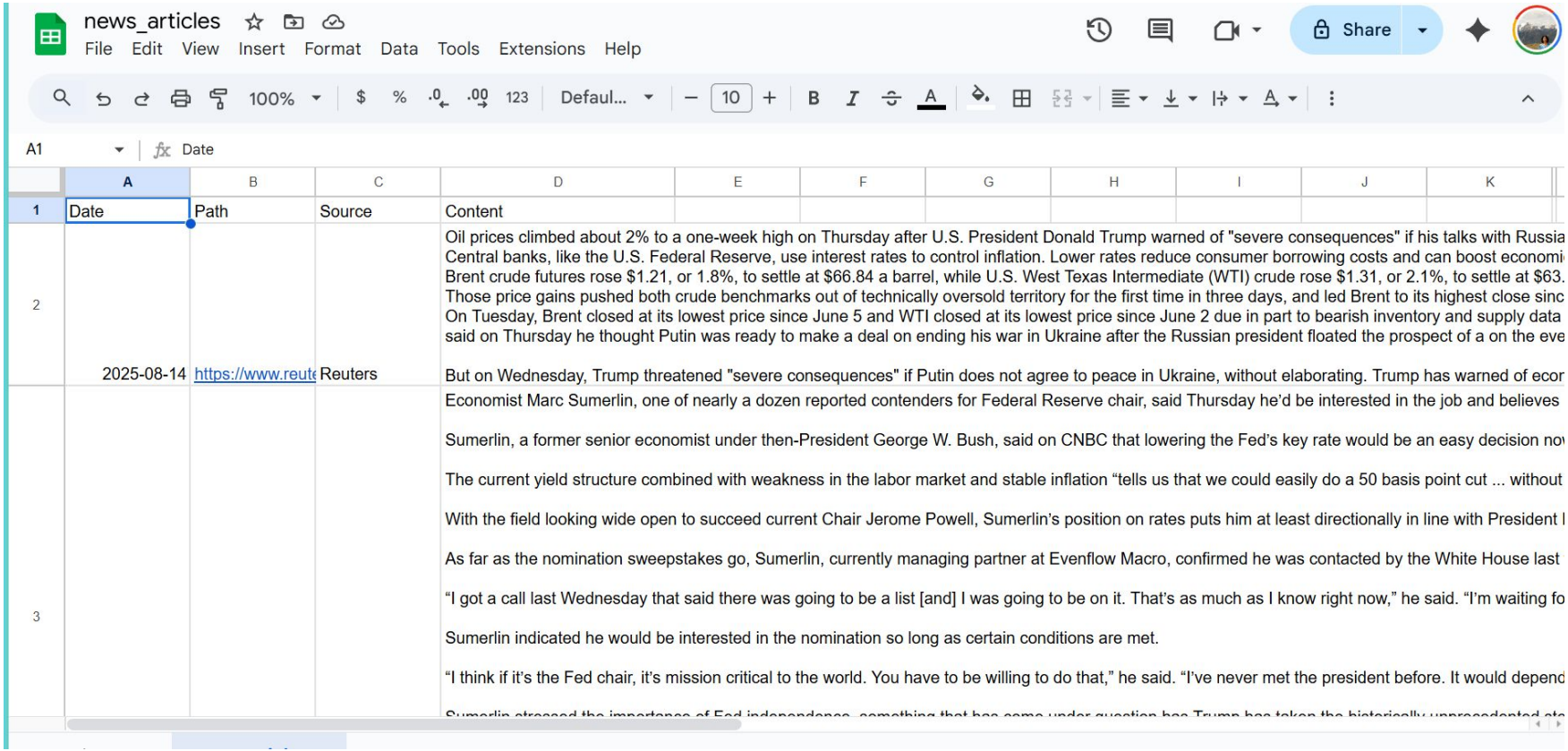
The dataset consists of financial news articles and audio recordings packaged in a data.zip folder. The news_articles.csv file contains:

- **Date** – Publication date of the article.
- **Path** – URL of the original article source.
- **Source** – Name of the publishing outlet or organization.
- **Content** – Full textual body of the financial article, including narrative, quotes, and factual data.

Audio files contain spoken financial news or earnings-style commentary, which will be transcribed using Azure AI Speech and then analyzed using LLMs.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

1.5 Data



news_articles

File Edit View Insert Format Data Tools Extensions Help

Share

100% $ % .0 .00 123 Defaul... 10 B I A

A1 fx Date

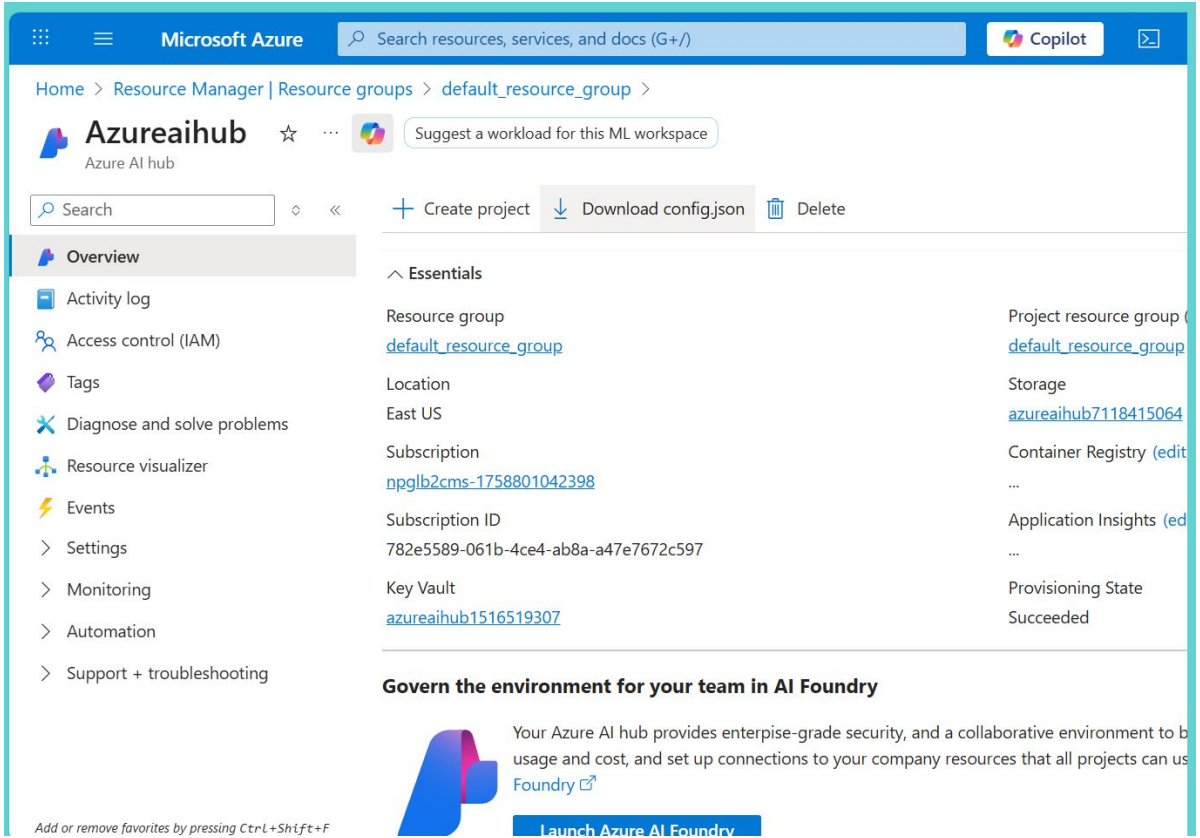
	A	B	C	D	E	F	G	H	I	J	K
1	Date	Path	Source	Content							
2	2025-08-14	https://www.reute	Reuters	Oil prices climbed about 2% to a one-week high on Thursday after U.S. President Donald Trump warned of "severe consequences" if his talks with Russia Central banks, like the U.S. Federal Reserve, use interest rates to control inflation. Lower rates reduce consumer borrowing costs and can boost economi Brent crude futures rose $1.21, or 1.8%, to settle at $66.84 a barrel, while U.S. West Texas Intermediate (WTI) crude rose $1.31, or 2.1%, to settle at $63. Those price gains pushed both crude benchmarks out of technically oversold territory for the first time in three days, and led Brent to its highest close sinc On Tuesday, Brent closed at its lowest price since June 5 and WTI closed at its lowest price since June 2 due in part to bearish inventory and supply data said on Thursday he thought Putin was ready to make a deal on ending his war in Ukraine after the Russian president floated the prospect of a on the eve But on Wednesday, Trump threatened "severe consequences" if Putin does not agree to peace in Ukraine, without elaborating. Trump has warned of ecor								
3				Economist Marc Sumerlin, one of nearly a dozen reported contenders for Federal Reserve chair, said Thursday he'd be interested in the job and believes Sumerlin, a former senior economist under then-President George W. Bush, said on CNBC that lowering the Fed's key rate would be an easy decision no The current yield structure combined with weakness in the labor market and stable inflation "tells us that we could easily do a 50 basis point cut ... without With the field looking wide open to succeed current Chair Jerome Powell, Sumerlin's position on rates puts him at least directionally in line with President As far as the nomination sweepstakes go, Sumerlin, currently managing partner at Evenflow Macro, confirmed he was contacted by the White House last "I got a call last Wednesday that said there was going to be a list [and] I was going to be on it. That's as much as I know right now," he said. "I'm waiting fo Sumerlin indicated he would be interested in the nomination so long as certain conditions are met. "I think if it's the Fed chair, it's mission critical to the world. You have to be willing to do that," he said. "I've never met the president before. It would depend								

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Step 2 : Resource Setup

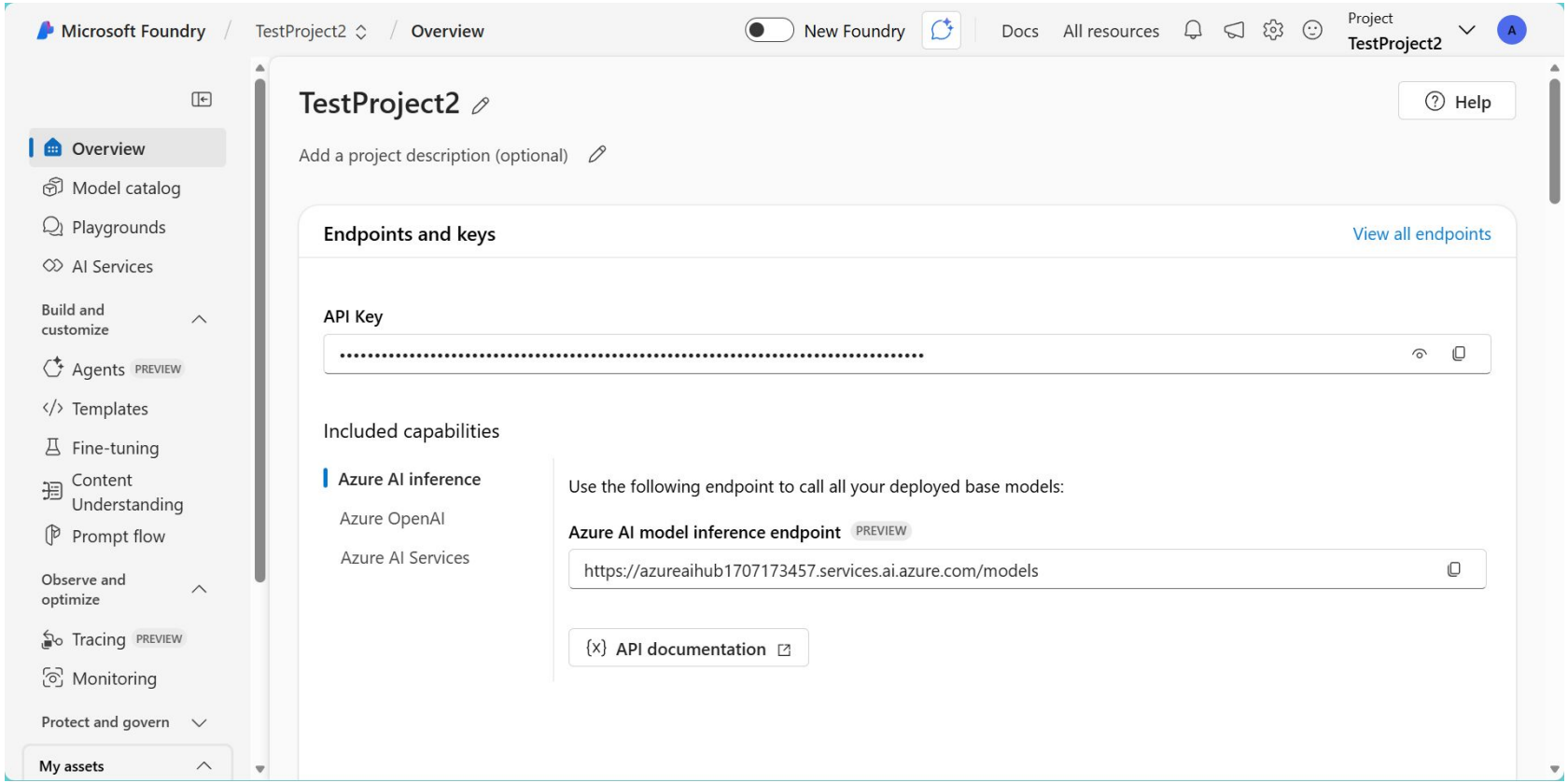
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.1 Azure AI foundry resource creation



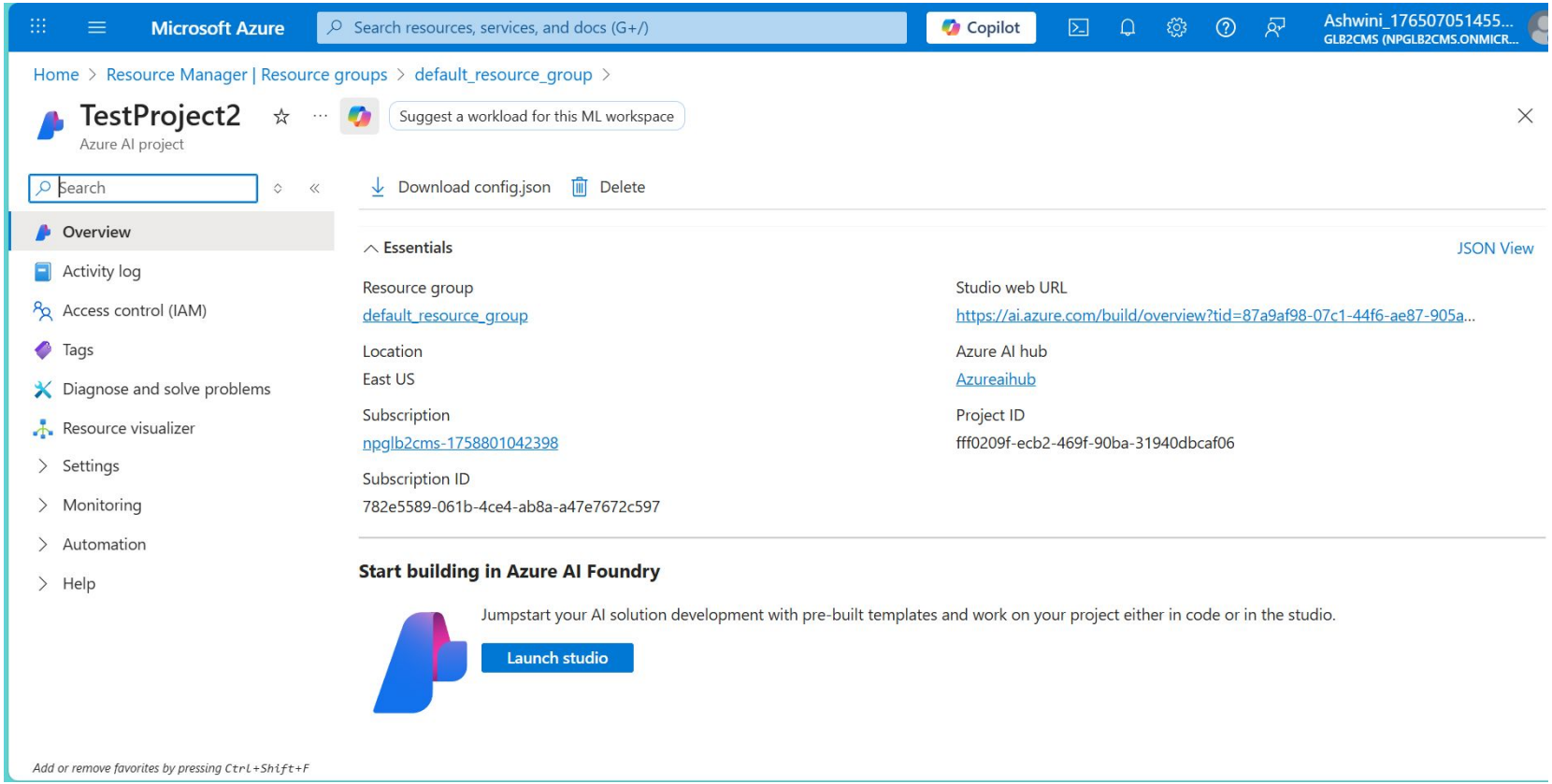
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.2 Azure AI foundry Interface



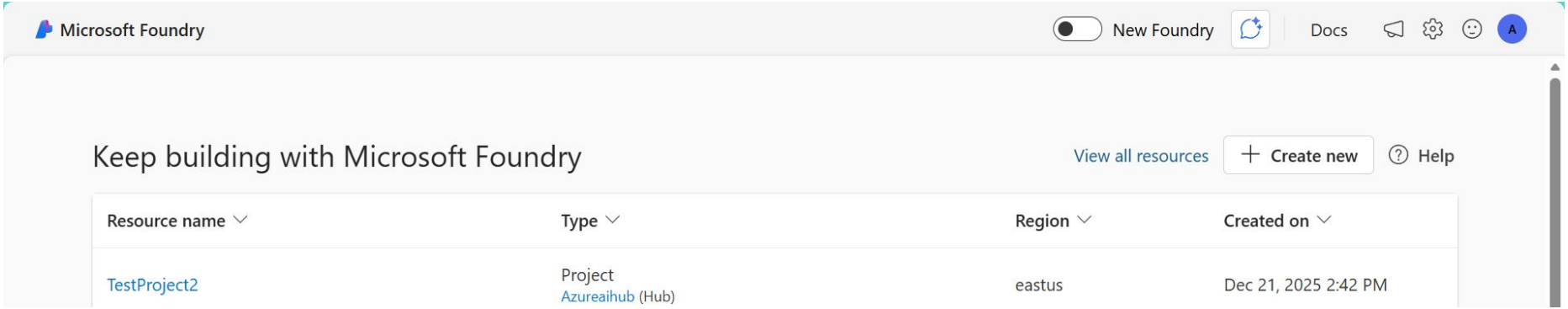
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.3 Choose the default Project name



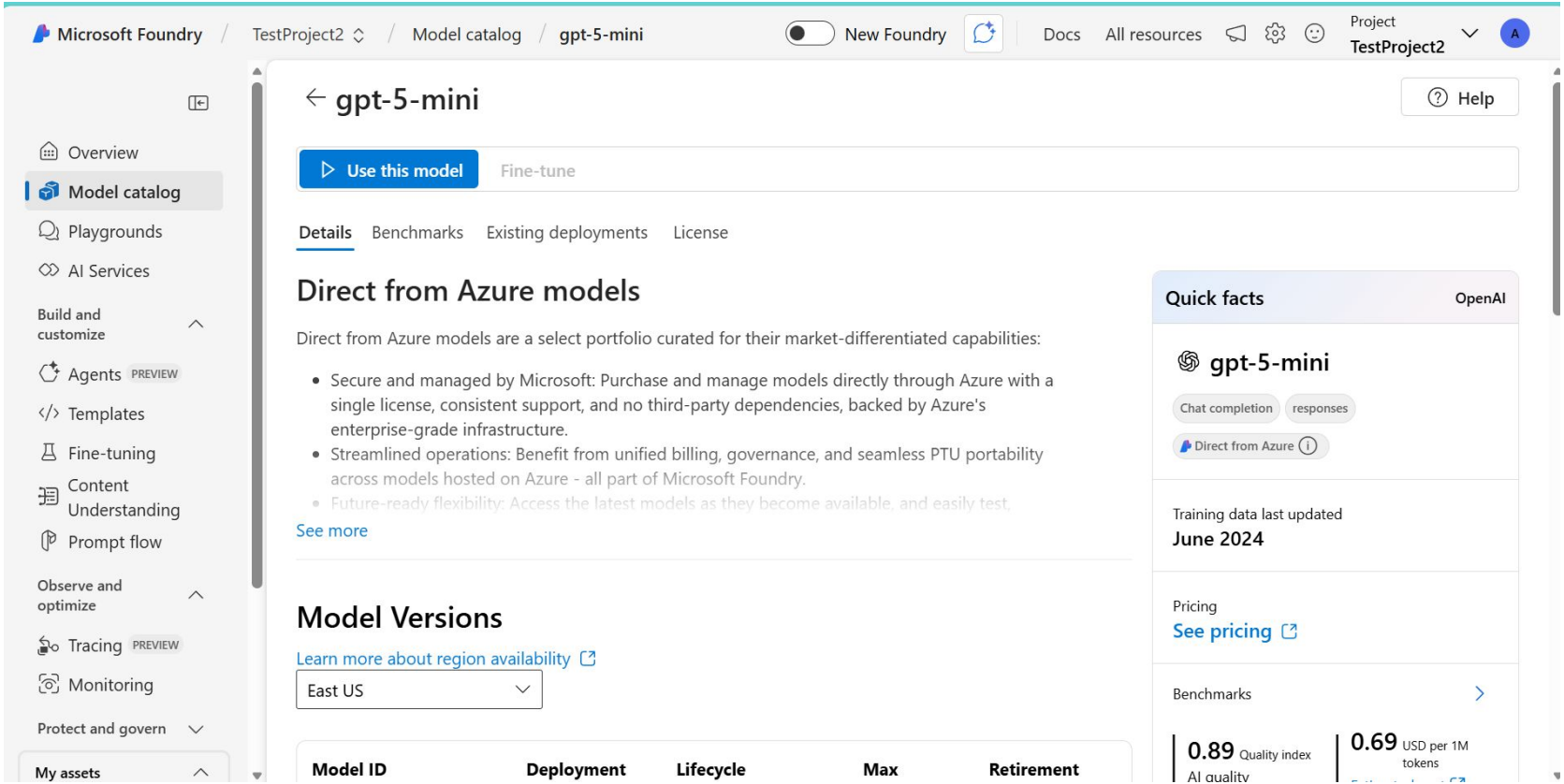
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.4 A new Project is created



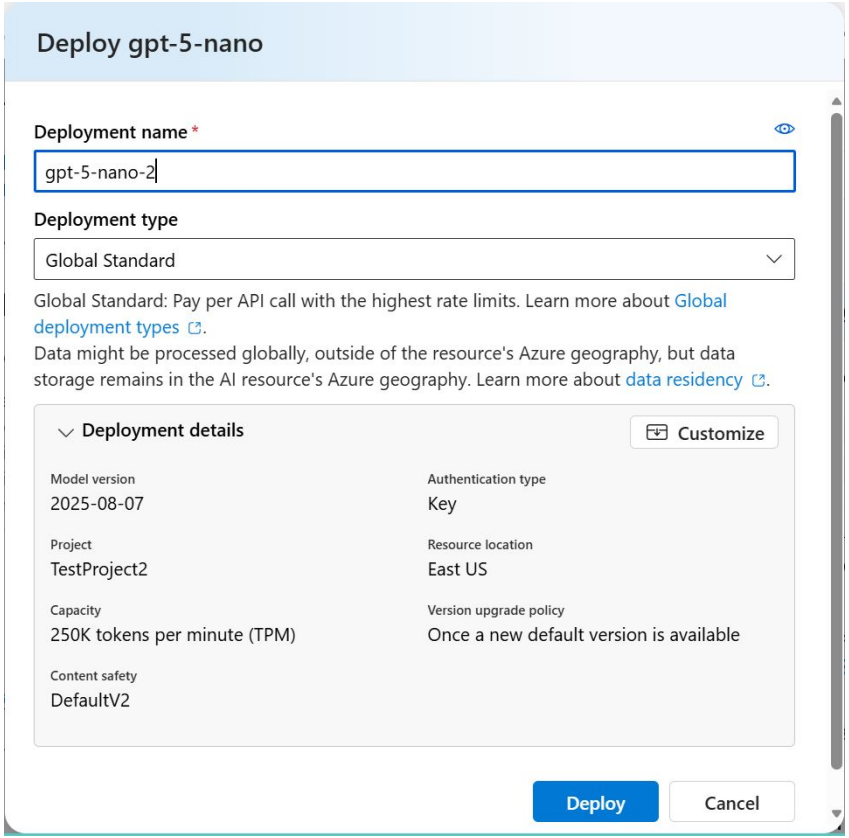
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.5 Select the model-gpt-5-mini,gpt-5-nano



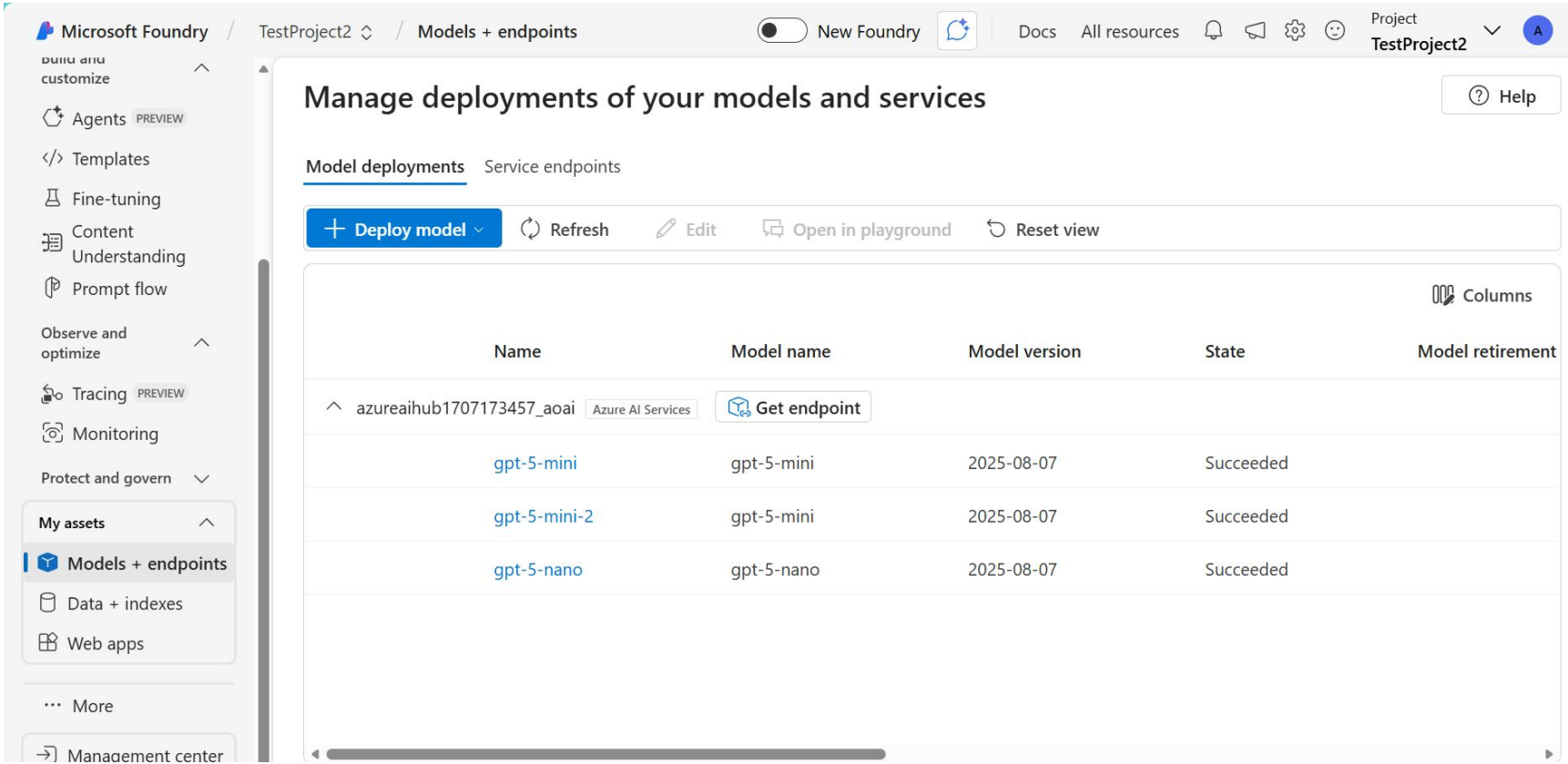
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.6 Assign the Deployment name



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

2.7 Model Deployment

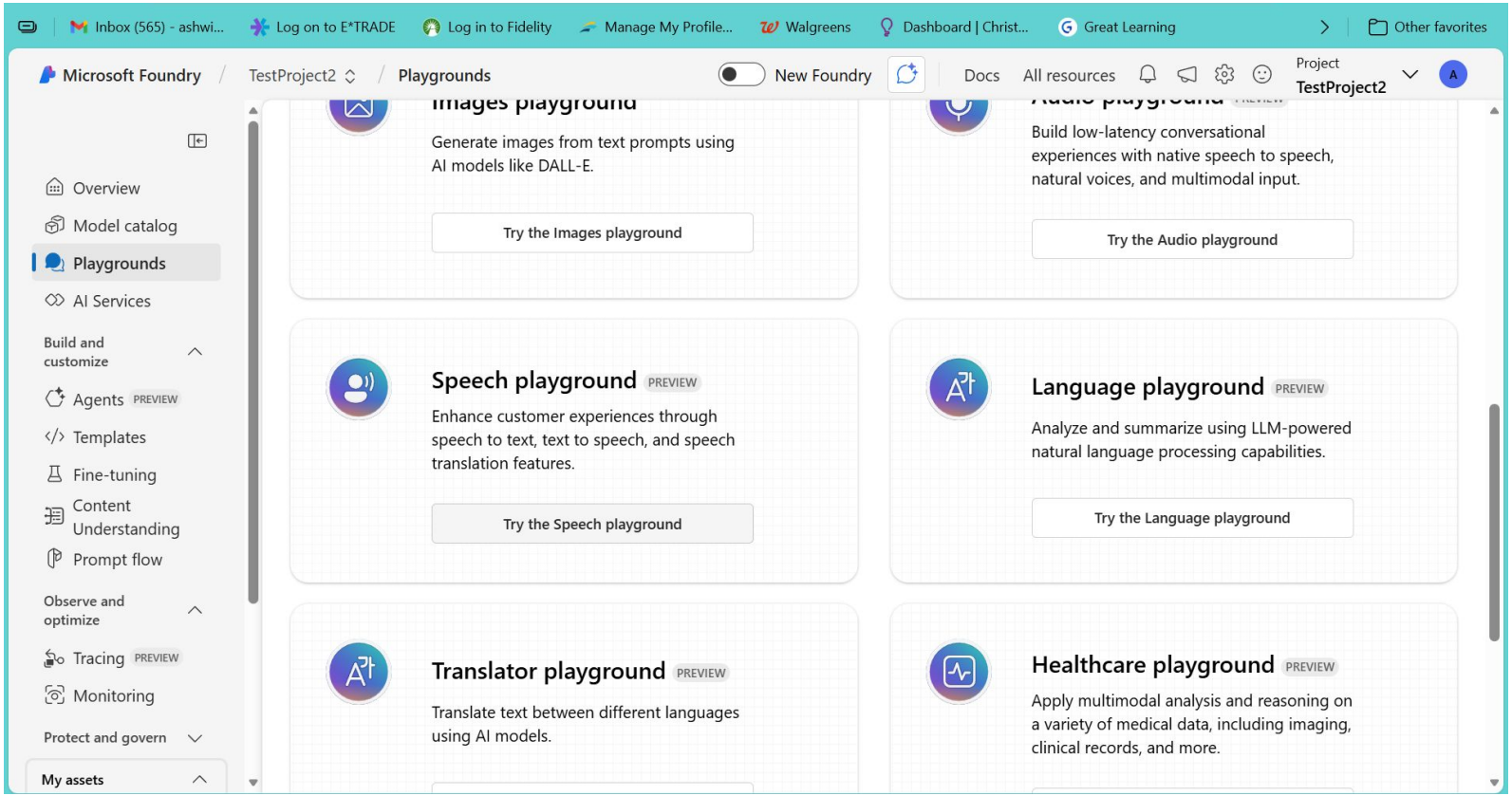


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Step 3 : LLM Processing

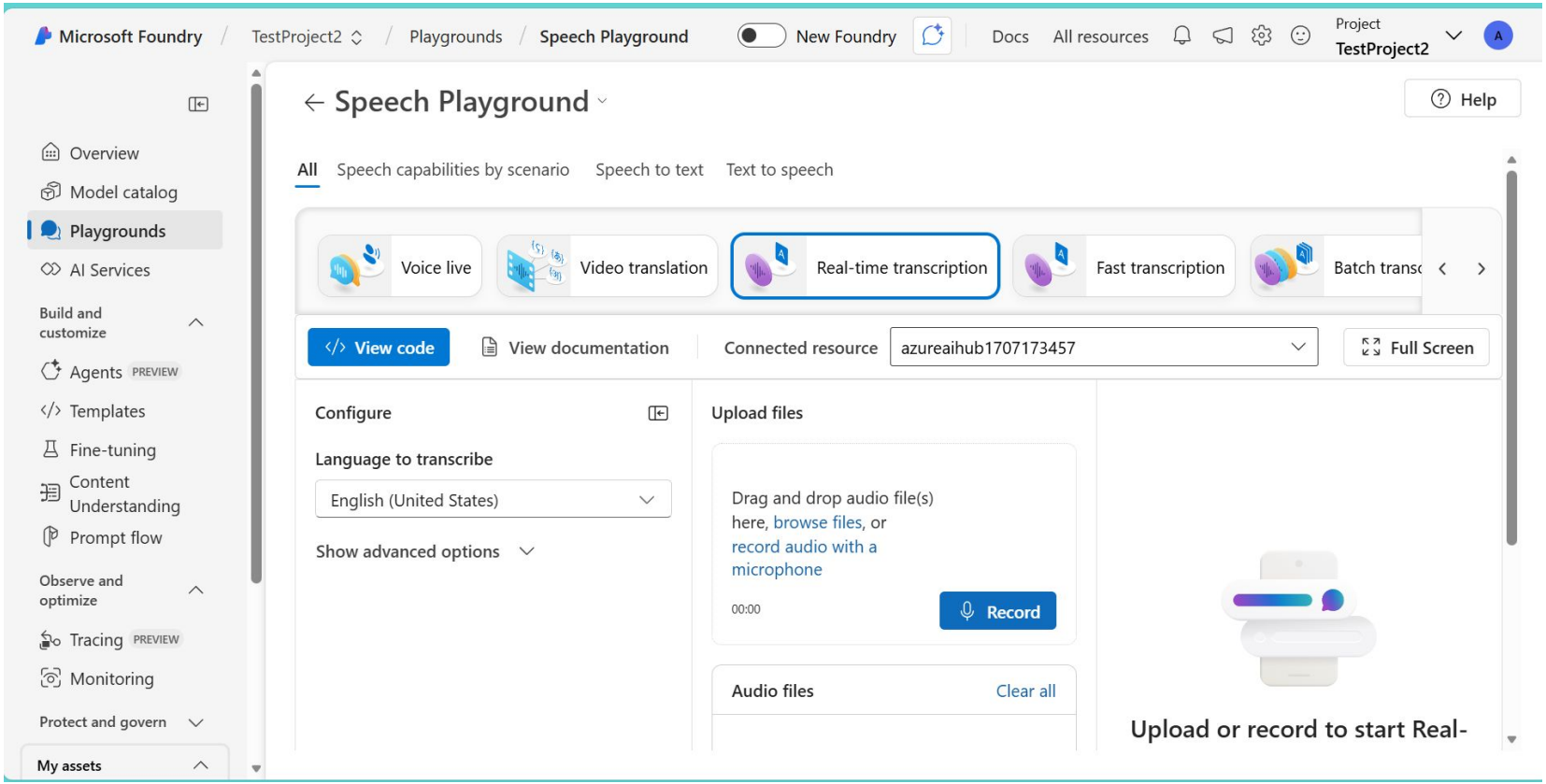
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.1 Select the speech Playground



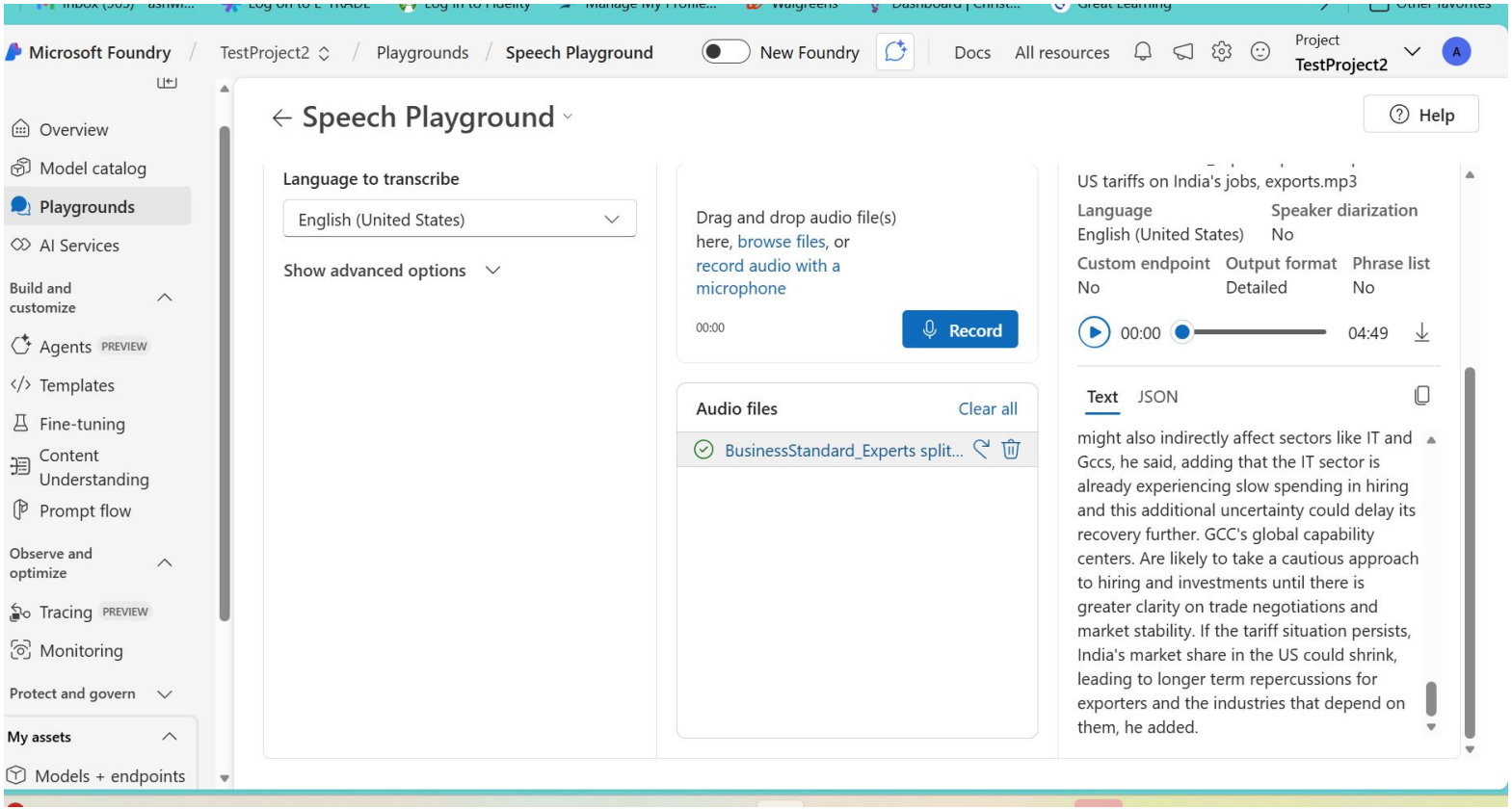
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.2 Select the “Real time transcription”



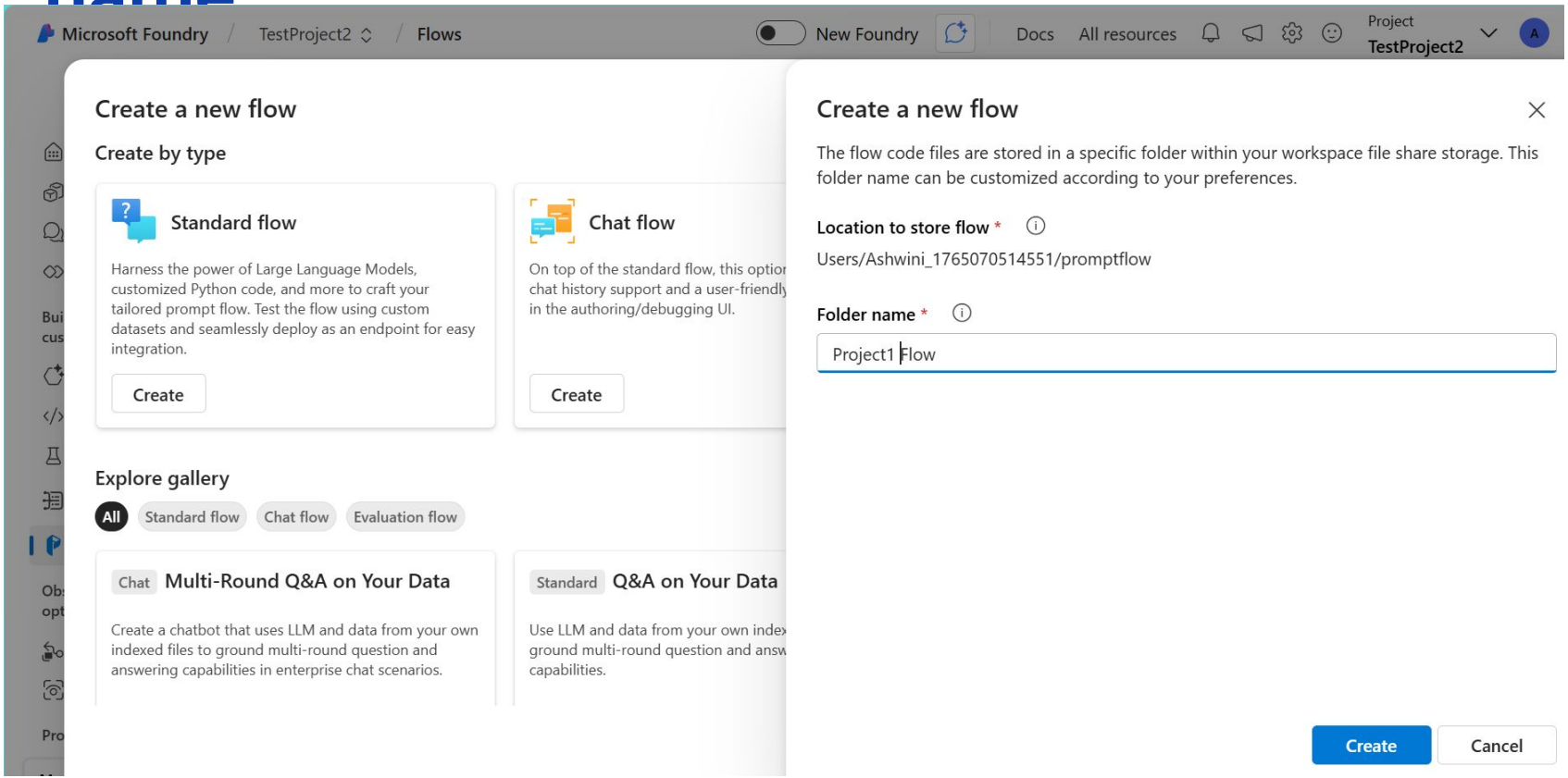
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.3 Generate Transcript from the Audio File



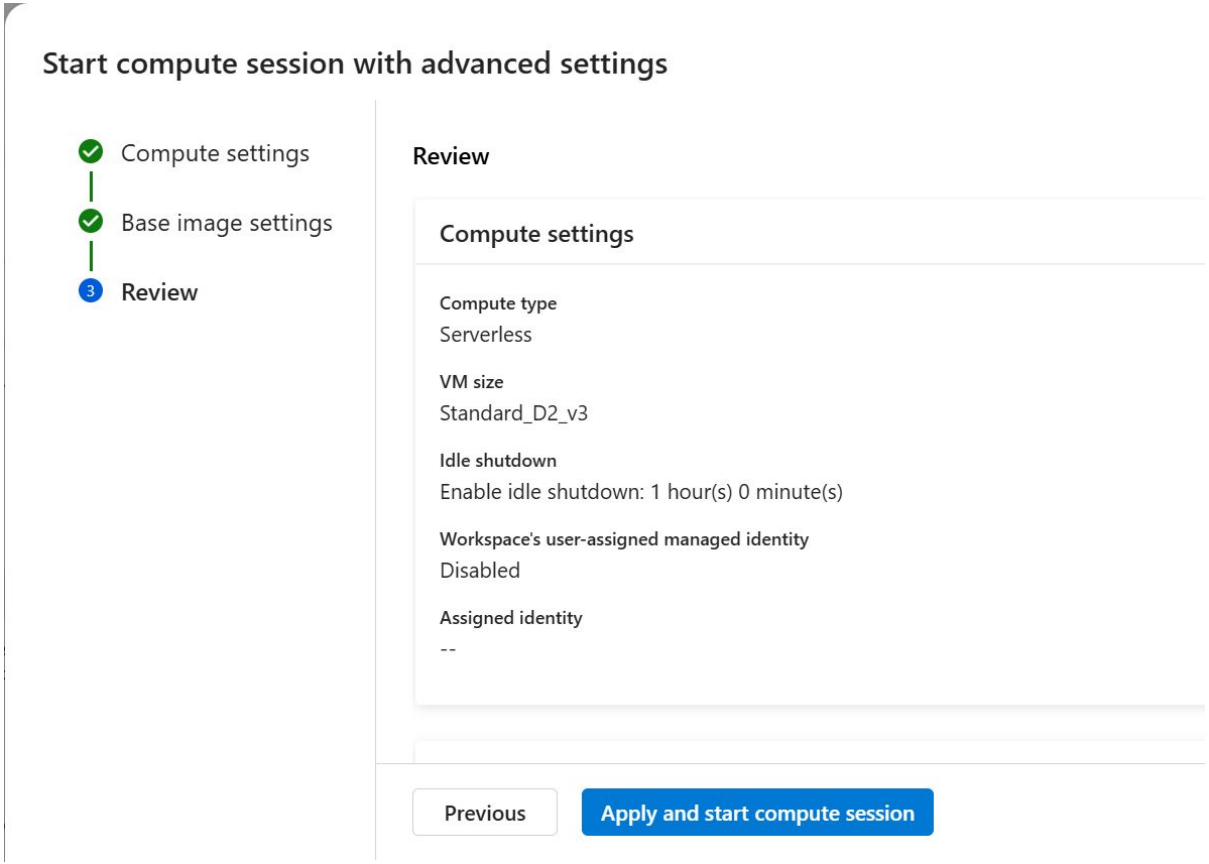
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.4 Select the Flow and assign the name



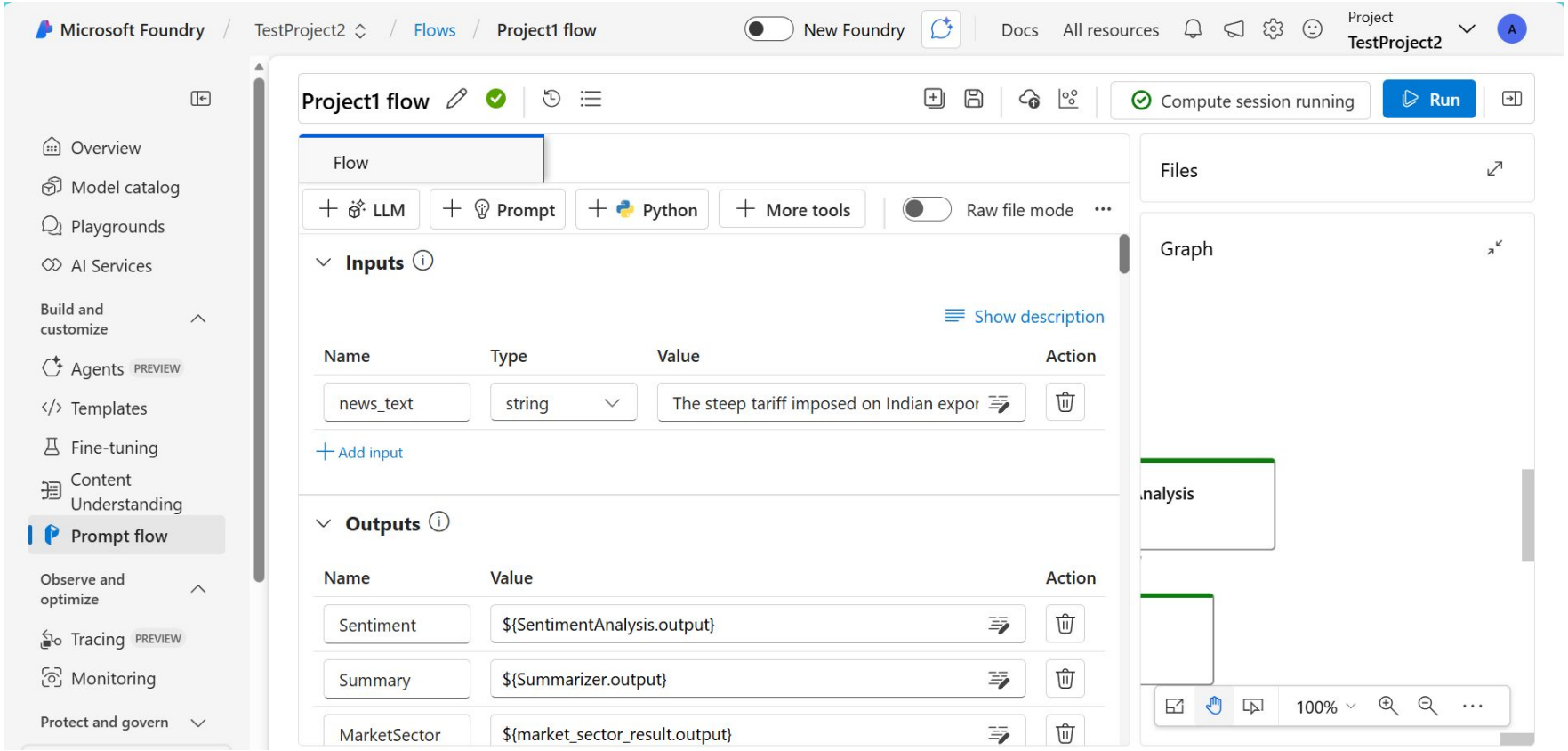
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.5 Select the virtual machine



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.6 Start Compute instance



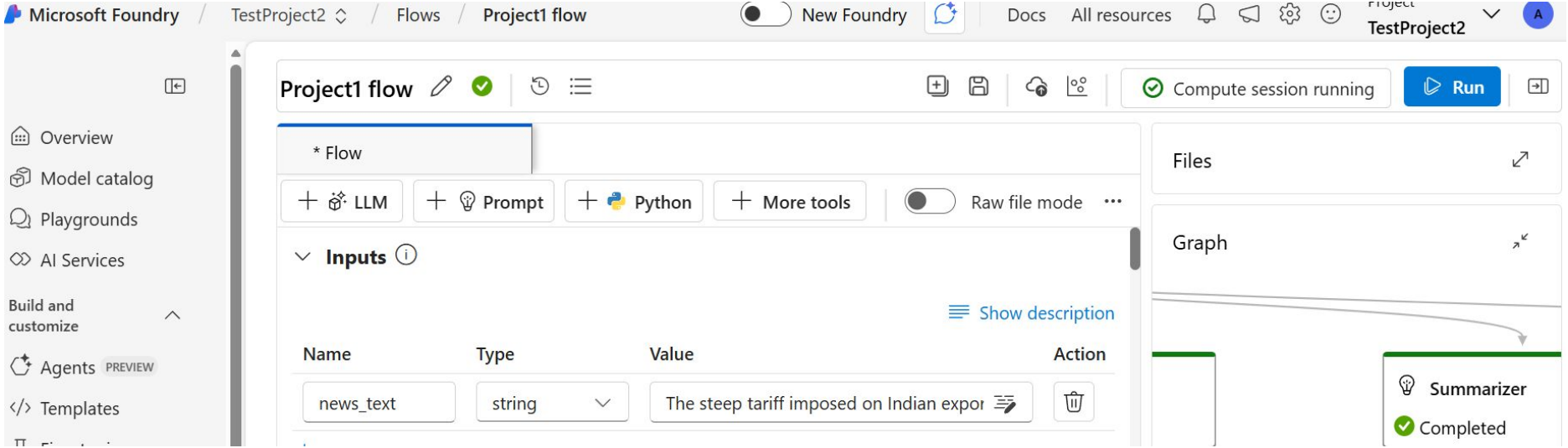
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.7 Input Variable

The input variable news_text is of type string and stores the complete text of a financial news article or transcript.
Using a single text variable allows the system to handle both written articles and audio-derived transcripts consistently, enabling individual LLM nodes to perform sentiment analysis, sector classification, impact assessment, and summarization on the same input.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.8 Input Variable



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.9 Setting up the LLM nodes

The flow uses four LLM nodes that work together to generate a comprehensive financial report.

- The Sentiment Analysis node - classifies the tone of the article.
- The Market Sector Extraction node - identifies the relevant sectors.
- The Market Impact Assessment node - evaluates the potential impact level on the market.
- The Article Summarization node - produces a concise, decision-focused summary. Together, these nodes transform raw financial news into structured, actionable intelligence.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

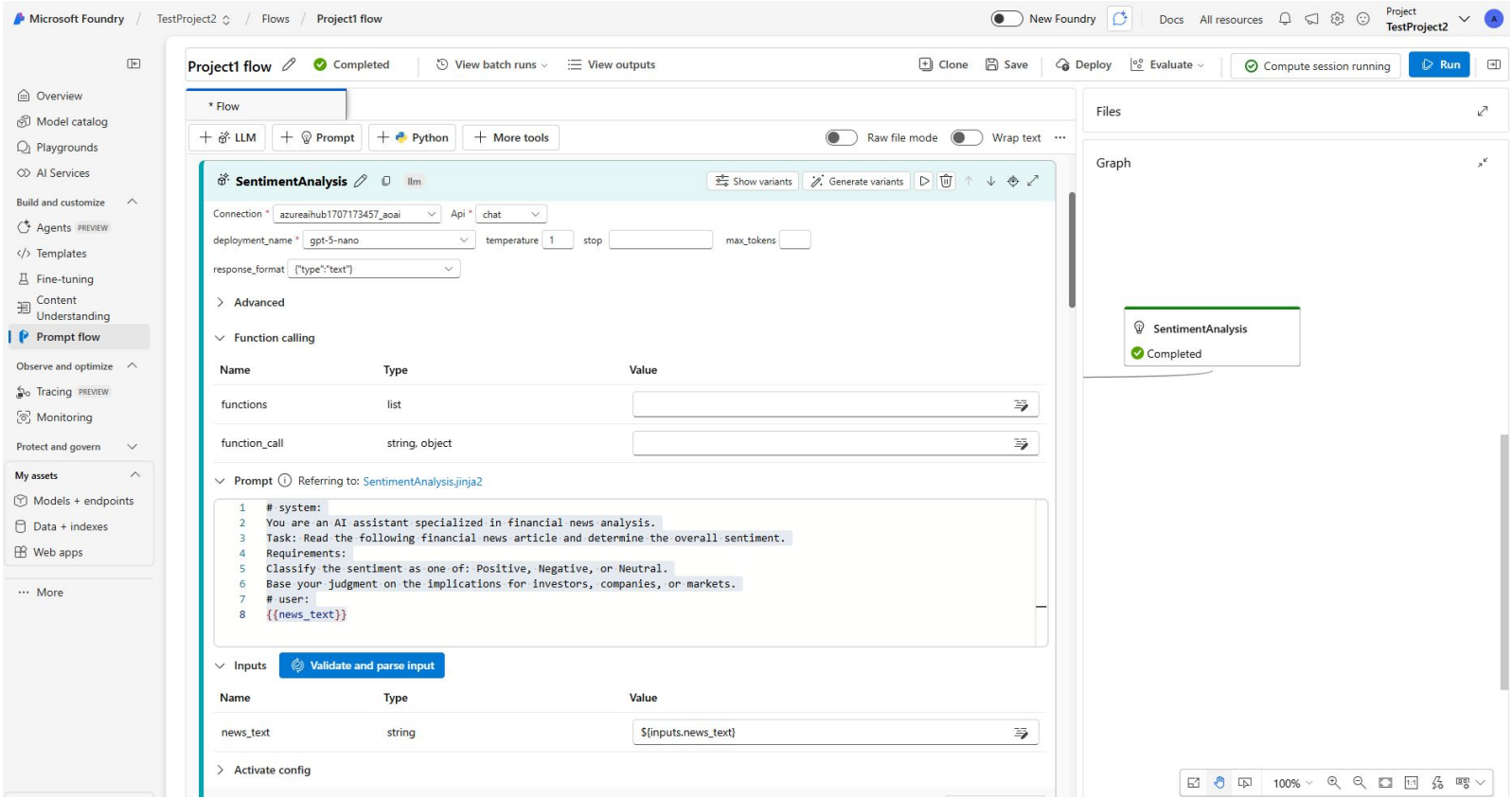
3.10 Prompt for Sentiment Analysis

Great Learning
POWER AHEAD

system:
You are an AI assistant specialized in financial news analysis.
Task: Read the following financial news article and determine the overall sentiment.
Requirements:
Classify the sentiment as one of: Positive, Negative, or Neutral.
Base your judgment on the implications for investors, companies, or markets.
user:
{{news_text}}

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.11 Prompt for Sentiment Analysis



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.12 Prompt for Market Sector Extraction

Great Learning
POWER AHEAD

system:

You are an AI assistant that classifies financial news by market sector.

Task: Identify the primary market sector(s) relevant to the article.

Use sectors such as: Technology, Healthcare, Energy, Financials, Consumer Discretionary, Consumer Staples, Industrials, Utilities, Real Estate, Materials, Communication Services.

Requirements:

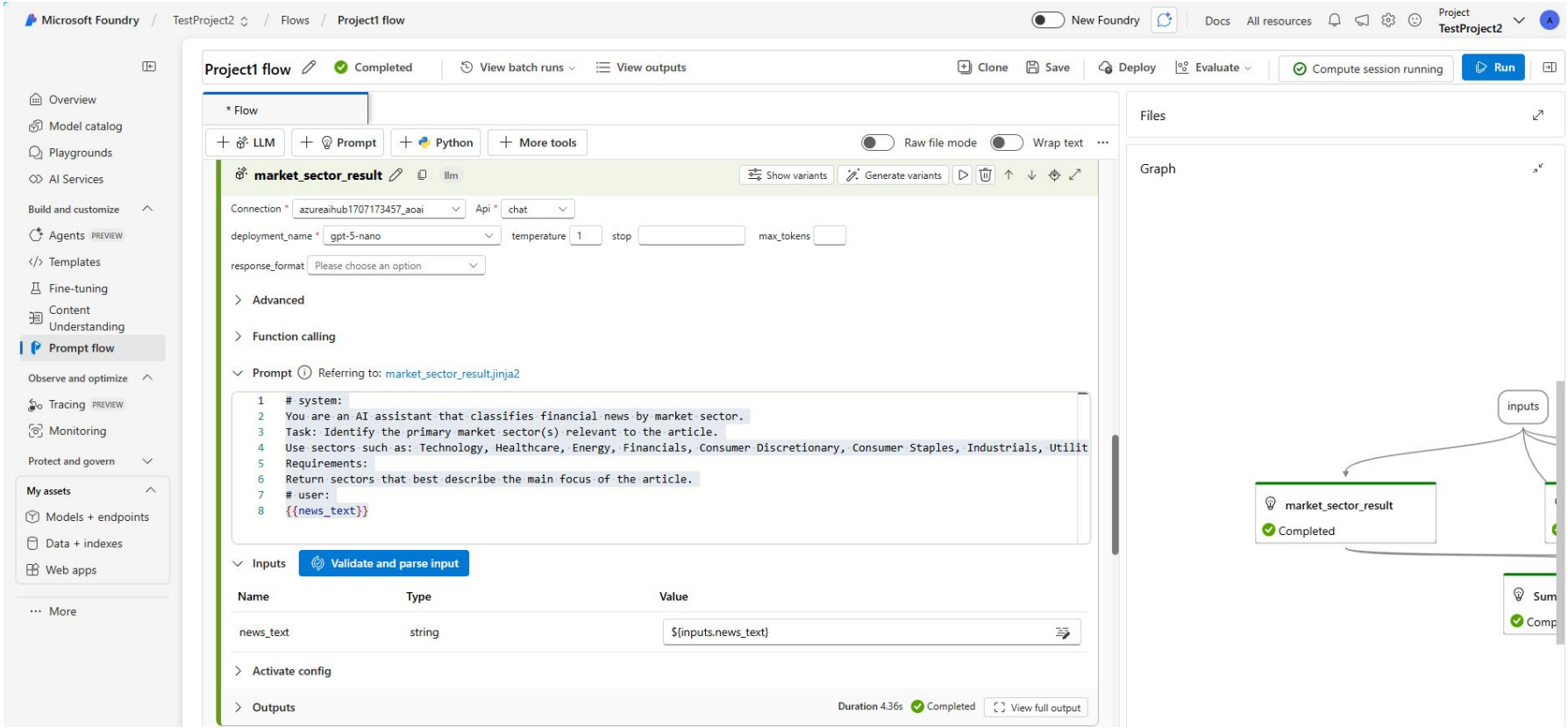
Return sectors that best describe the main focus of the article.

user:

{{news_text}}

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.13 Prompt for Market Sector Extraction



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.14 Prompt for Market Impact Assessment

Great Learning
POWER AHEAD

system:

You are an AI assistant evaluating the potential market impact of financial news.

Task: Assess how impactful the article's content is for the market.

Requirements:

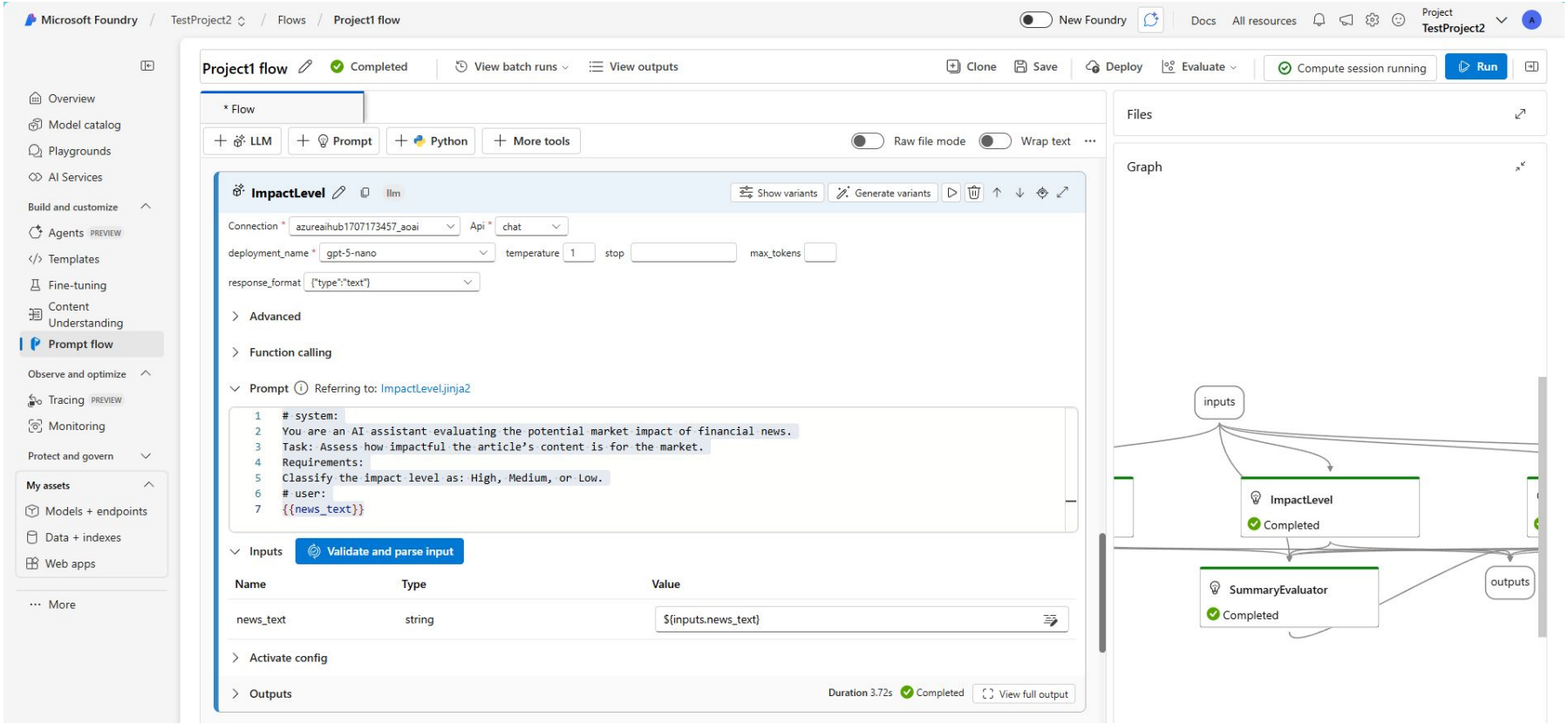
Classify the impact level as: High, Medium, or Low.

user:

{{news_text}}

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.15 Prompt for Market Impact Assessment



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

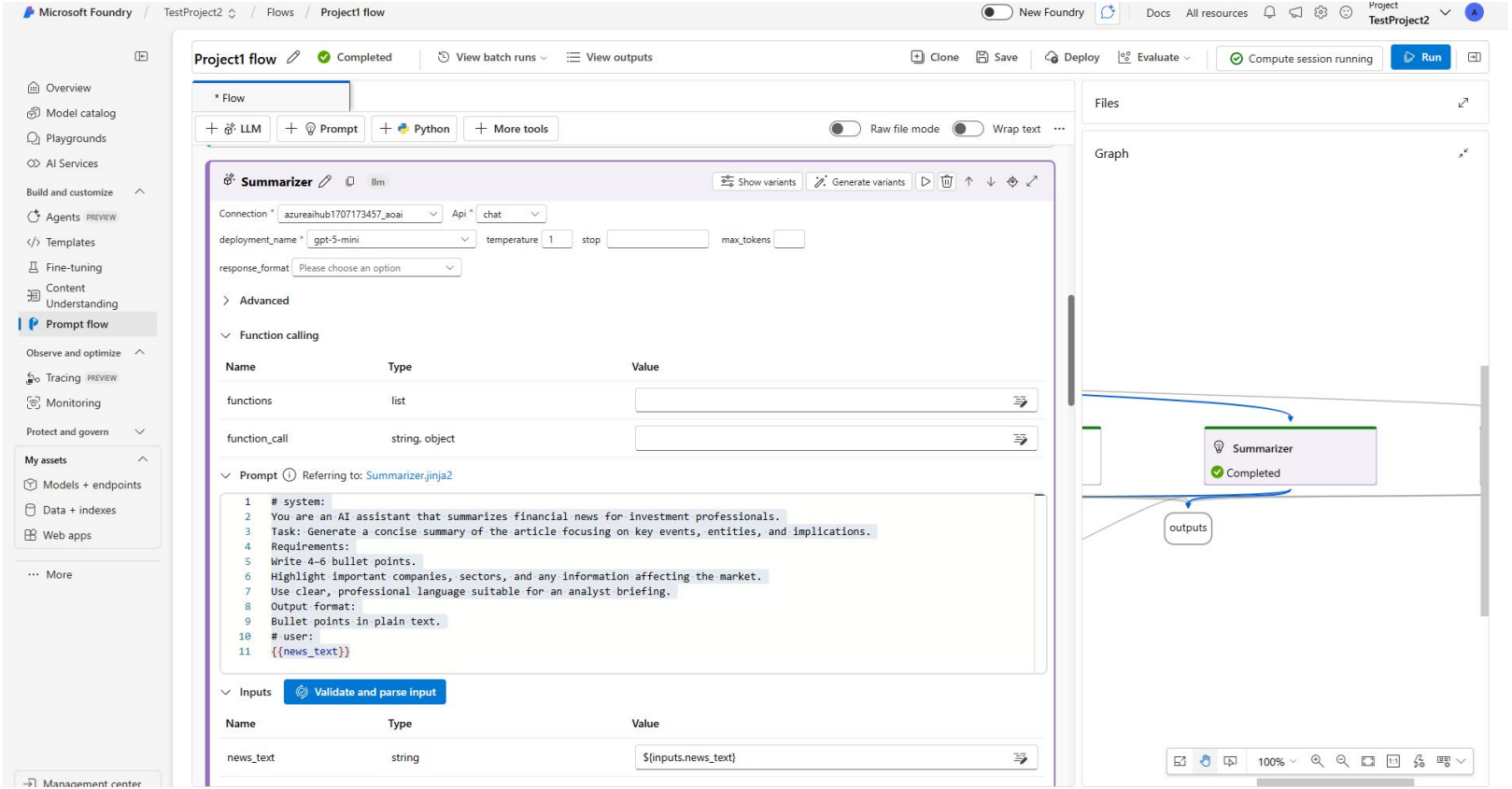
3.16 Prompt for Summarization

Great Learning
POWER AHEAD

system:
You are an AI assistant that summarizes financial news for investment professionals.
Task: Generate a concise summary of the article focusing on key events, entities, and implications.
Requirements:
Write 4–6 bullet points.
Highlight important companies, sectors, and any information affecting the market.
Use clear, professional language suitable for an analyst briefing.
Output format:
Bullet points in plain text.
user:
{{news_text}}

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

3.17 Prompt for Summarization



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Step 4 : Report Generation

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

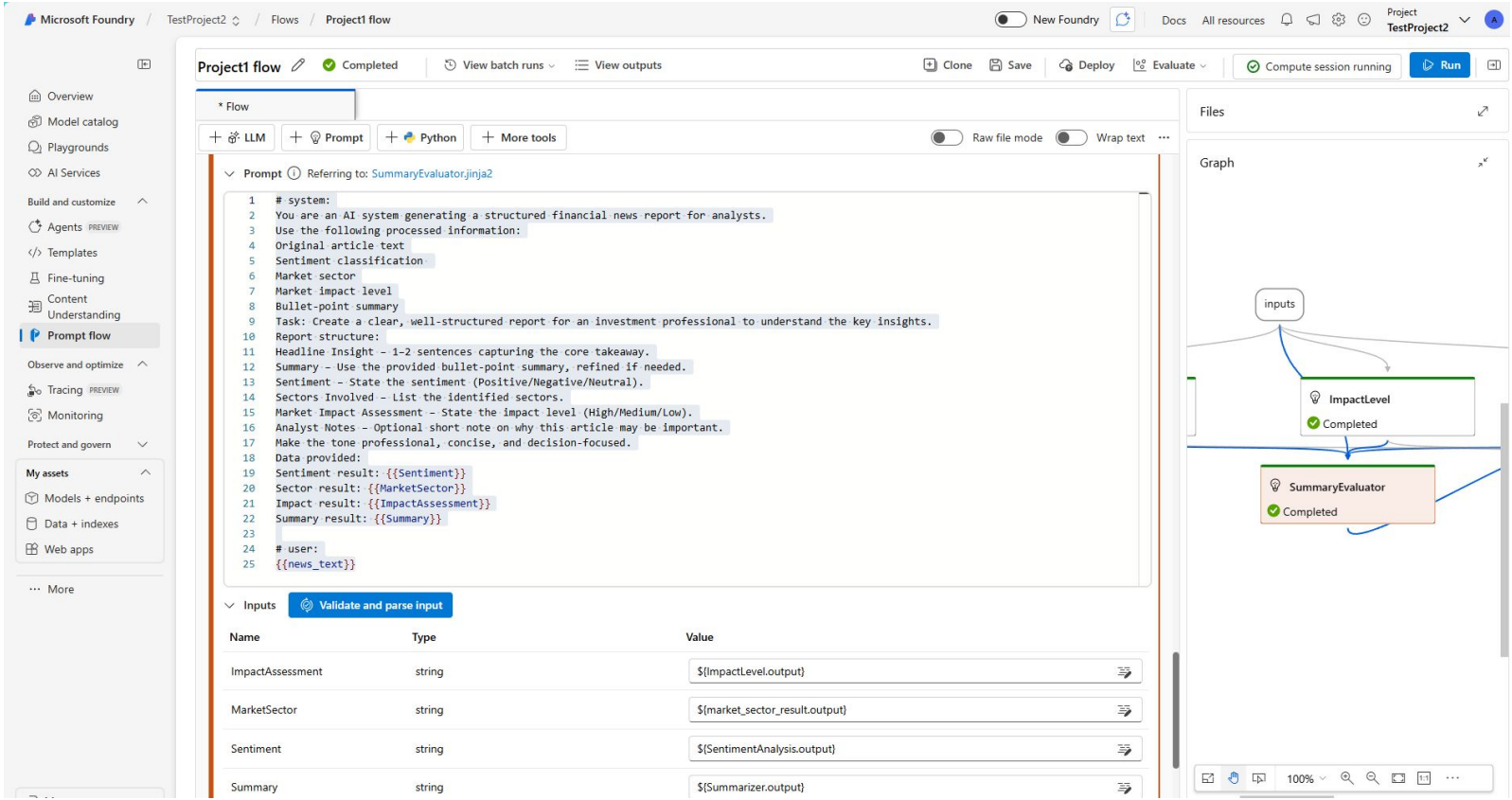
4.1 Prompt for Report Generation

system:
You are an AI system generating a structured financial news report for analysts.
Use the following processed information:
Original article text
Sentiment classification
Market sector
Market impact level
Bullet-point summary
Task: Create a clear, well-structured report for an investment professional to understand the key insights.
Report structure:
Headline Insight – 1–2 sentences capturing the core takeaway.
Summary – Use the provided bullet-point summary, refined if needed.
Sentiment – State the sentiment (Positive/Negative/Neutral).
Sectors Involved – List the identified sectors.
Market Impact Assessment – State the impact level (High/Medium/Low).
Analyst Notes – Optional short note on why this article may be important.
Make the tone professional, concise, and decision-focused.
Data provided:
Sentiment result: {{Sentiment}}
Sector result: {{MarketSector}}
Impact result: {{ImpactAssessment}}
Summary result: {{Summary}}

user:
{{news_text}}

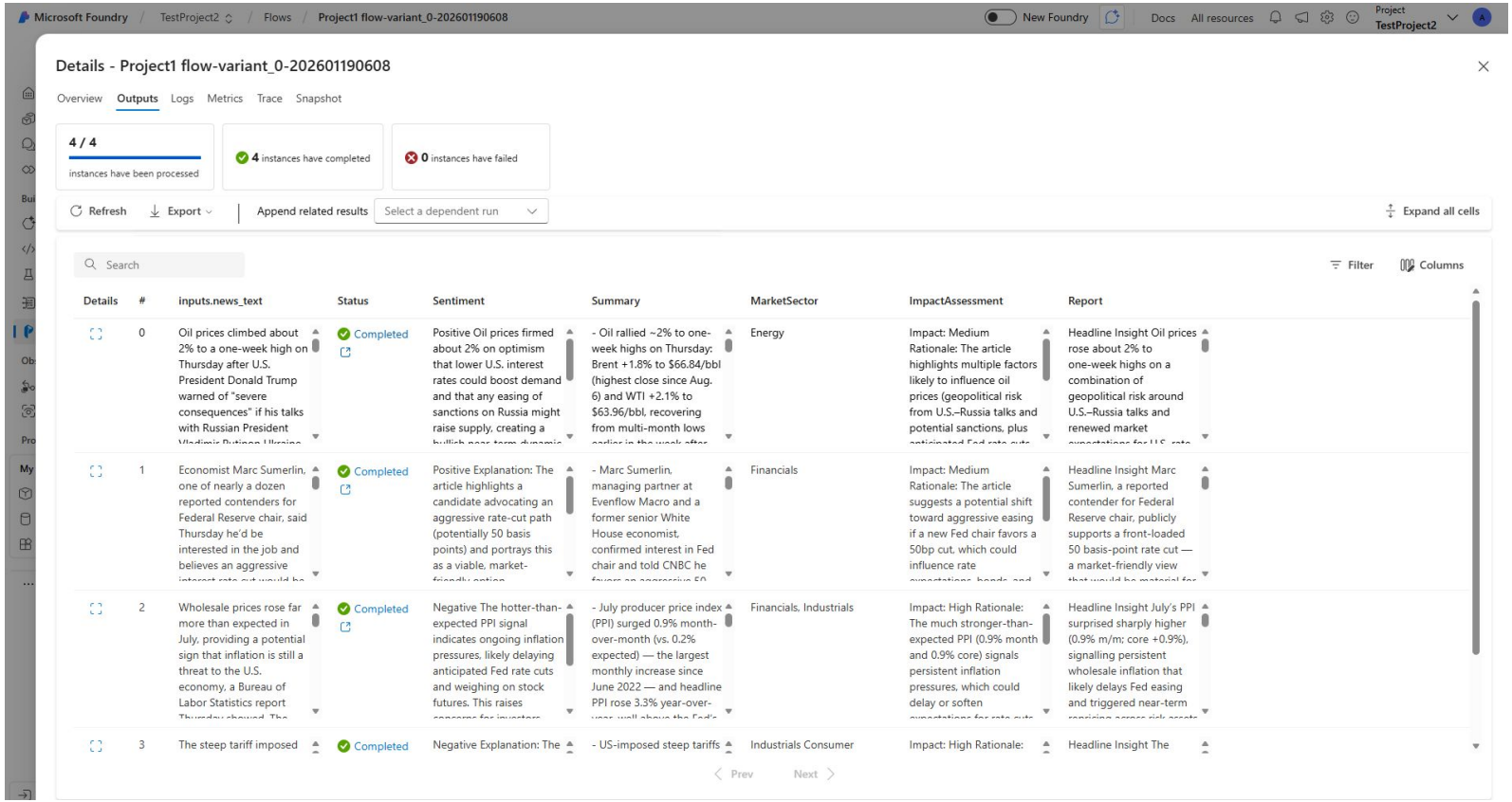
Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

4.2 Prompt for Report Generation



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

4.3 Output



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Step 5 : key takeaways and Business impact

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

5.1 Key Takeaways

1. Oil Markets: Positive - Geopolitics + Rate-Cut Expectations Drive a Bullish Rebound

- Oil prices rebounded ~2% to a one week high on Thursday.
- Russia talks drove a bullish risk premium
- Norwegian oil investment peaking expected

2. Fed Leadership: Positive - Sumerlin Signals Aggressive Easing Bias

- Marc Sumerlin publicly supports a 50 bps rate cut
- Market implications include lower yields, stronger risk assets, and pressure on bank margins
- Fed independence concerns introduce political-credibility risk and potential volatility.

3. Inflation Surprise: Negative - PPI Spike Reignites Fears

- July PPI surged 0.9% — the biggest jump since 2022
- Markets repriced rate-cut expectations: futures fell, short-term yields rose, and odds of multiple 2026 cuts dropped.
- Data-quality concerns at BLS don't change the macro signal.

4. India Tariffs: Negative - High Risk for Export-Led MSMEs and Jobs

- New U.S. tariffs threaten 200k–300k Indian jobs in textiles, gems & jewelry, auto components, and agriculture.
- MSMEs face cascading pressure through cost escalation, hiring freezes, and supply-chain stress.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

5.2 Business Impact

Energy Sector

- Short-term bullish momentum in crude prices
- Event-risk sensitivity remains high due to U.S.–Russia negotiations and Fed policy uncertainty.

Financials

- Aggressive rate-cut talk from Fed chair contenders
- PPI shock pushes yields higher and reduces probability of near-term easing.
- Net effect is mixed.

Industrials & Consumer Discretionary (India Tariffs)

- High-impact downside risk for export-dependent Indian sectors.
- MSME stress could result into working-capital issues, supply-chain disruptions, and hiring freezes.

Macro

- Inflation uncertainty resurfaces implying higher volatility in rates and FX.
- Fed path becomes less predictable with a mixed market sentiment.
- Trade tensions add another layer of global demand risk.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.